

Language-Model Aircraft Concepts from Pre-1899 Sources: A Flight-Mechanics Audit

Ehsan Roohi*

University of Massachusetts Amherst, Amherst, Massachusetts 01003

Three large language models were asked to design a piloted, powered airplane from a short packet of aeronautical sources published before 1899, and their proposals were revised through archived feedback rounds. This paper reports an independent evaluation in which every model statement was classified by its level of evidence and, where possible, checked by recomputing mass properties, trimming the lifting surfaces with a vortex-lattice method, comparing required with assumed propulsive power, and computing static and rate stability derivatives. The initial concepts failed in different ways: a tandem-wing design was statically unstable in pitch, one biplane could trim only by moving its tail through its fin, and another trimmed with a small static margin but needed slightly more power than its assumed engine target. After repeated feedback, including an evaluator-proposed braced tail that one model adopted, the remaining biplane showed surface-model static margins of 4–10% of the reference chord across eight loading states, but its heavy-case power reserve fell to 11% when its induced drag was computed rather than assumed. No design reached installed six-component trim, validated dynamic modes, or demonstrated structural adequacy. Unlike the Wright brothers’ test-driven development, the process yielded auditable concepts but no flight evidence.

Nomenclature

a_D	=	assumed moment arm of the added drag increment, m
a_T	=	thrust-line moment arm, $z_G - z_{\text{hub}}$, m
b	=	reference span, m
C_D	=	drag coefficient
$C_{D,\text{ind}}$	=	induced-drag coefficient predicted by the lifting-surface model
ΔC_D	=	added drag coefficient specified by the evaluator
C_L, C_{L_α}	=	lift coefficient; lift-curve slope, rad^{-1}
C_{L_q}	=	lift derivative with respect to $\hat{q}$

*Lecturer, Department of Mechanical and Industrial Engineering; roohie@umass.edu.

C_l, C_n	=	rolling- and yawing-moment coefficients
C_{l_β}, C_{n_β}	=	rolling- and yawing-moment derivatives with respect to β , rad^{-1}
C_{l_p}, C_{n_r}	=	roll- and yaw-damping derivatives with respect to $\hat{p}$ and $\hat{r}$
C_m	=	pitching-moment coefficient about the center of gravity, nose-up positive
C_{m_α}, C_{m_q}	=	pitching-moment derivatives with respect to α (rad^{-1}) and $\hat{q}$
c_{ref}	=	reference chord, m
D	=	drag, N
g	=	gravitational acceleration, 9.80665 m/s^2
k	=	tail half-span scale factor
L, L_j	=	total lift; lift of lifting surface j , N
M_{aero}, M_T	=	aerodynamic and thrust-line pitching moments about the center of gravity, N·m
m, m_i	=	aircraft mass; mass of ledger item i , kg
P_e	=	engine output power, W
P_{shaft}	=	power delivered to the propeller shafts, W
p, q_p, r	=	roll, pitch, and yaw rates, rad/s
$\hat{p}, \hat{q}, \hat{r}$	=	nondimensional rates, $pb/2V, q_p c_{\text{ref}}/2V, rb/2V$
q	=	dynamic pressure, $\rho V^2/2$, Pa
R_x, R_z	=	forward and upward force residuals, N
R_m	=	nose-up pitching-moment residual about the center of gravity, N·m
$\mathbf{r}_i, \mathbf{r}_G$	=	position of ledger item i ; center-of-gravity position, m
S	=	reference area, m^2
SM	=	fixed-control static margin, $-C_{m_\alpha}/C_{L_\alpha}$, fraction of c_{ref}
T	=	thrust along the thrust line, N
V	=	airspeed, m/s
W	=	weight, mg , N
x, y, z	=	body axes: aft, right, and up positive, m
x_G, z_G	=	center-of-gravity coordinates, m
x_N	=	fixed-control neutral-point station, m
z_{hub}	=	propeller hub height, m
α	=	angle of attack of the thrust/body reference line, deg or rad
β	=	sideslip angle, rad
δ	=	pitch-control command, deg

η_g	=	assumed engine-to-shaft drive efficiency
η_p	=	assumed propulsive efficiency, DV/P_{shaft}
ρ	=	air density, 1.225 kg/m^3

Subscripts

0	=	value printed in a model response
h	=	evaluator hybrid estimate
s	=	evaluator lifting-surface solution

I. Introduction

LARGE language models (LLMs) are increasingly used to generate and refine aircraft concepts. Recent studies have compared prompting strategies for structured concept generation [1], examined LLM assistance in aircraft-system design and simulation coding [2], embedded LLMs in agentic conceptual-design workflows [3, 4], and linked design descriptions to geometry and simulation artifacts [5]. These efforts use different representations and endpoints, and none is a single measure of “LLM aircraft design.” A common difficulty is that an LLM produces fluent text and convincing drawings together. In conceptual design, however, a configuration is credible only when its mass, lift, pitching moments, power, control travel, and structure close at a common operating point, and when each change is carried back through the dependent estimates [6–10]. Iterative self-feedback can improve generated text [11, 12], but it does not reliably correct reasoning without external information [13]. Generated aircraft therefore need independent, calculation-based evaluation, and the evaluation must keep a model’s claims separate from the evaluator’s results.

The history of powered flight offers a demanding reference case for such an evaluation. By the end of 1898, published work by Cayley, Lilienthal, Chanute, and others already treated equilibrium, propulsion, materials, and operation as coupled problems [14]. The Wright brothers’ subsequent route to the 1903 Flyer was a sequence of tests and revisions rather than a single drawing: disappointing lift from their 1900 and 1901 gliders led them to measure wing sections in their own wind tunnel, to revise the 1902 glider, and to develop three-axis control before installing an engine and two propellers [15–17]. Modern reconstructions quantify that achievement. Full-scale tests of the 1901 glider [18], analyses of the 1902 glider [19, 20], and wind-tunnel tests of a 1903 replica at NASA Ames [21, 22] show that the Flyer was not passively stable. The Ames tests indicate a static margin of about -25% of the mean aerodynamic chord near 25 kt [22], and a linearized reconstruction gives a divergent pitch mode with a time to double of about 0.41 s [23]. The Flyer flew because its pilot stabilized it continuously [24]. The 1905 Flyer III, which first made sustained, repeatable flights, remained statically unstable but had larger control moment arms, more pitch damping, and a slower divergence, with a time to double of about 2.2 s [23, 25, 26]. Studies of Lilienthal’s gliders published in this journal follow the same chain from historical interpretation through reconstruction and measurement to constrained flight evidence [27–30].

These works suggest an evidence standard for any claim that a generated design “could fly”: geometry and mass must be reconstructible, trim must be admissible for the actual control mechanism, propulsion must be matched to drag at a stated operating point, and stability must be judged together with pilot or augmentation authority. They also show that a positive static margin is not the relevant historical benchmark. The relevant benchmark is the process by which tests exposed false assumptions and recorded design changes followed.

This paper reports an evaluation of three LLM-generated aircraft against that standard. Each model received the same short packet of pre-1899 sources and a structured design brief, and its proposal was then revised through case-specific feedback over a series of archived rounds. The contributions are as follows:

- 1) An evaluation protocol that assigns every model statement an evidence level and passes each archived design through a fixed sequence of mass, trim, power, and stability gates.
- 2) A traceable record of how feedback changed three generated aircraft, which changes survived independent checks, and where the evaluator itself intervened in the design.
- 3) Quantitative trim, static and rate derivative, loading-envelope, structural-mass, and power-margin results for the designs that reached each gate.
- 4) A comparison with the Wright brothers’ development that separates passive static stability, pilot-controlled flight, and evidence-producing iteration.

The study does not claim that any design is flightworthy, does not rank the underlying models, and does not test historical inference independent of the models’ training data.

Section II describes the design experiment and the evidence levels. Section III presents the evaluation methods. Sections IV–VI report the baseline concepts, the intermediate feedback rounds, and the refinement of the final candidate. Section VII discusses the results in relation to the Wright development, and Sec. VIII states the conclusions. Detailed tables, superseded results, and full component keys are given in the Supplemental Material.

II. Design Experiment

A. Source Packet and Design Brief

The common design date was 31 December 1898. The input packet contained four short paraphrases with bibliographic links: Cayley’s 1809–1810 work on aerial navigation, Lilienthal’s account of his 1893 experiments as reproduced by Chanute [14], Chanute’s concluding discussion, and an 1892 chapter on wings and uncertain power estimates. The packet was not a complete collection of period literature, and the date restriction applied to the supplied sources, not to the models’ training. The experiment is therefore historically constrained prompting, not a test of what a reader in 1898 could have inferred.

The brief asked for an original, engine-powered aircraft carrying a human operator and capable in principle of

Table 2 Design phases, archived versions, and principal outcomes

Phase	Versions	Request content	Principal outcome
A. Concept	V0	Source packet and design brief	Three architectures; internal contradictions in mass and geometry
B. Clarification	V1	Non-overlapping mass rows, centroids, envelopes, control axes, interfaces	Baseline audit: Astra statically unstable; Fable trim blocked by tail–fin conflict; Opus trims with 3% power shortfall
C. First feedback	V2	V1 results returned as labeled evidence; revised geometry, mass, controls, and propulsion requested	Astra rearchitected; Opus revised; Fable geometry unresolved
D. Integration	V3, V3b, V5–V8-R, V10, V11a/b, V12, V13	Integrated mass, control, propulsion, and dynamics plans; defect corrections; installation geometry	Ledger and power-chain errors exposed; Astra and Opus on hold; Fable power chain short by 0.52 kW
E. Refinement	V14–V16	Installation and loading feedback; evaluator tail concept supplied in V16 request	Fable surface-model margins 4–10% over eight loading states; power reserve sensitive to drag

repeated directed flight and safe return. It requested an engineering article of approximately 1500–2200 words containing physical reasoning, surface geometry and section ordinates, a dimensioned component-coordinate table, force and moment considerations, disturbance response, propulsion and mass estimates, launch and landing reasoning, unknowns, and proposed experiments. Layout and control architecture were left open. The models were required to disclose suspected recall of later knowledge and were prohibited from claiming real experiments.

The recorded endpoint identifiers were `gpt-6-astra`, `claude-fable-5-1`, and `claude-opus-5-5`, referred to below as Astra, Fable, and Opus. These identifiers are experimental labels, not capability rankings. Output limits and retries differed between models (Sec. II.B), so the study supports no comparison of model capability.

B. Response Versions and Feedback Phases

Table 2 summarizes the archived response versions. Labels follow the archive and are not contiguous. The first complete response to each request was retained; truncated responses were archived separately and never merged with retries. Requests after V1 contained modern evaluator results and constraints, so later versions are researcher-assisted revisions rather than independent historical inferences. In phase E, the evaluator also generated one design change, an enlarged and braced horizontal tail for Fable, to test whether the stability deficit could be closed; that change is reported as an evaluator intervention and is not credited to the model.

Output budgets were unequal. In phase B, Astra and Fable completed within a 32,000-token output limit, whereas Opus required a separately recorded 64,000-token attempt. In phase D, two V3 responses were truncated at 20,000 tokens and completed in independent 48,000-token retries. In phase E, Fable’s final request required a compact structured response after earlier attempts ended in truncation, a timeout, and an invalid request, and Opus’s first V12 response was truncated and later replayed in full. One phase-D request (V11a), which supplied only hashes of earlier responses, was a protocol error: all three models correctly declined to act on it, and it is not counted as a design failure.

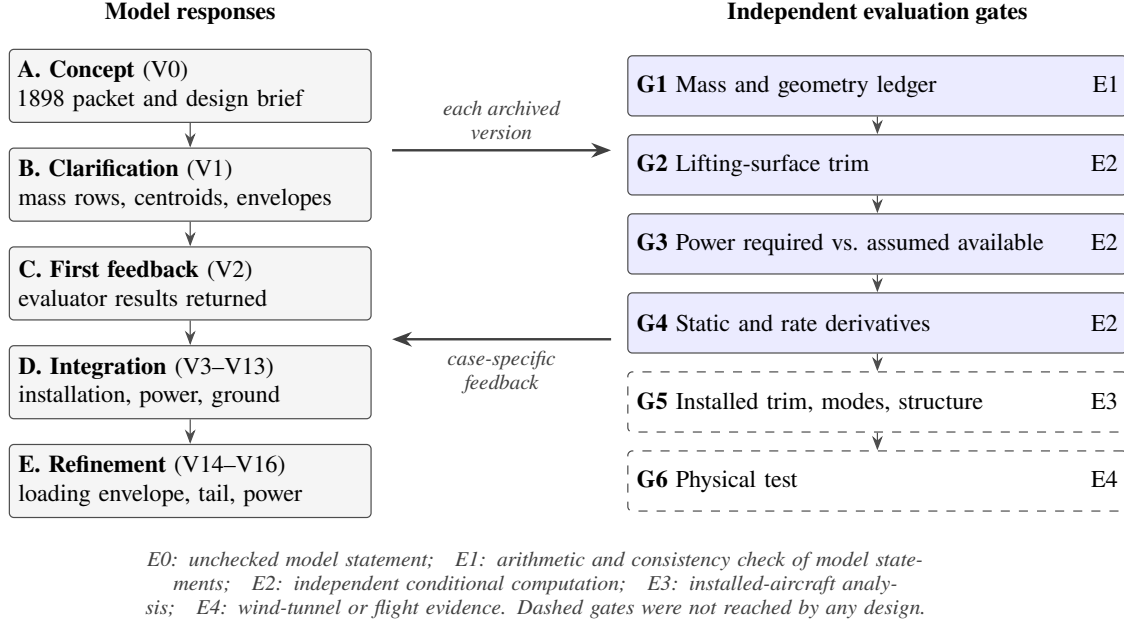


Fig. 1 Study workflow: archived model responses (left) and independent evaluation gates (right).

C. Evidence Levels and Evaluation Gates

Every quantitative statement was assigned one of five evidence levels: E0, an unchecked model statement; E1, a model statement whose arithmetic and internal consistency were verified; E2, an independent but conditional evaluator computation; E3, an installed-aircraft analysis including body, propulsion, measured or validated section data, and complete inertia; and E4, wind-tunnel or flight evidence. Figure 1 shows how each archived version passed through gates G1–G6. Gates G1–G4 were applied wherever the response contained enough information. No design reached G5 or G6, so no result in this paper exceeds level E2.

III. Evaluation Methods

A. Coordinates, Mass Properties, and Geometry

All analyses use body axes with x aft, y to the operator’s right, and z up. Astra’s forward-positive coordinates were reflected into this convention. Each design keeps its own origin, so x_G values cannot be compared across designs. Mass totals and centroids were recomputed from non-overlapping ledger rows,

$$m = \sum_i m_i, \quad \mathbf{r}_G = \frac{1}{m} \sum_i m_i \mathbf{r}_i, \quad W = mg, \quad (1)$$

and the moment reference was placed at the recomputed center of gravity (CG). A correct sum verifies arithmetic only; component masses remain model assumptions. Control-surface clearance was checked separately from aerodynamics by rigidly rotating the declared surface envelope about its hinge, whereas the aerodynamic model represents a control

command by rotating panel normals. Keeping the two operations separate prevents an aerodynamic trim from being accepted at a mechanically impossible control setting.

B. Aerodynamic Model

The lifting surfaces were modeled with the AVL vortex-lattice code, version 3.52 [31], which represents thin, rigid, attached-flow lifting surfaces [32, 33]. Body, pilot, landing-gear, bracing-wire, and propeller-slipstream loads, profile drag, ground effect, and elastic deformation were omitted. Section ordinates came from the model responses; aft tails were represented as flat plates where no section was specified. Baseline analyses used 8 and 12 chordwise by 40 and 80 spanwise panels per main half-surface; the later Fable analyses used 8 and 12 chordwise by 32 and 64 spanwise panels with matched fin stations. Two-mesh differences are reported as numerical sensitivity, not convergence or validation [34].

As a plausibility check, an untuned four-surface reconstruction of the 1903 Flyer’s main decks and canard was analyzed with the same code. Across four coarse–fine pairs, the largest changes were 1.06% in C_L , 1.81% in $C_{D,\text{ind}}$, and 0.00108 in C_m , whereas a change in assumed main-wing incidence altered C_L by 6.22–9.16%. Unresolved installation geometry, rather than discretization, therefore dominates the uncertainty of this class of model. The reconstruction’s moment reference at $0.3c_{\text{ref}}$ is not a sourced historical CG, so its pitch derivatives are not compared quantitatively with the generated designs.

C. Longitudinal Trim

For steady level flight with thrust along the body reference line, the forward and upward force residuals are

$$R_x = T \cos \alpha - D, \quad R_z = \sum_j L_j + T \sin \alpha - W, \quad (2)$$

where the sum includes tails carrying negative lift. Using the CG-referenced aerodynamic moment from the solver, the pitching-moment residual is

$$R_m = qSc_{\text{ref}}C_m + a_T T + a_D qS \Delta C_D, \quad D = qS (C_{D,\text{ind}} + \Delta C_D). \quad (3)$$

At a prescribed α , the control command δ was found from $R_m = 0$, and the speed and thrust then follow from the force equations:

$$q = \frac{W}{S [C_L + (C_{D,\text{ind}} + \Delta C_D) \tan \alpha]}, \quad V = \sqrt{\frac{2q}{\rho}}, \quad T = \frac{qS (C_{D,\text{ind}} + \Delta C_D)}{\cos \alpha}. \quad (4)$$

Equation (4) closes the force balance by construction: it yields the thrust that would be required, not evidence that an installed propeller can supply it. Control roots were interpolated only between computed commands, then re-evaluated

directly and accepted when $|R_x|/W$, $|R_z|/W$, and $|R_m|/(Wc_{\text{ref}})$ were all at most 10^{-3} . A root was also required to pass the clearance check of Sec. III.A.

D. Propulsion Power

Useful, shaft, and engine powers are related by

$$P_{\text{shaft}} = \frac{DV}{\eta_p}, \quad P_e = \frac{P_{\text{shaft}}}{\eta_g}, \quad P_{e,\text{min}} = \frac{1.15 DV}{\eta_p \eta_g}, \quad (5)$$

where $P_{e,\text{min}}$ includes the 15% power reserve used as a screen. In the baseline analysis, $\eta_p = 0.55, 0.50$, and 0.60 were taken from the Astra, Fable, and Opus responses, respectively. To test the sensitivity of a model's own power estimate to its induced drag, its printed induced-drag coefficient was replaced by the evaluator's surface solution at the same speed and mass while all other model assumptions were held fixed:

$$\eta_{p,0} = \frac{D_0 V}{P_0}, \quad D_h = D_0 + qS (C_{D,\text{ind},s} - C_{D,\text{ind},0}), \quad P_h = \frac{D_h V}{\eta_{p,0}}. \quad (6)$$

Equation (6) neither adds a second induced-drag estimate nor constitutes an engine or propeller map.

E. Static and Rate Derivatives

Fixed-control slopes were computed by perturbing α by ± 0.25 and ± 0.5 deg about each accepted trim state without retrimming. The static margin is

$$SM = -\frac{C_{m_\alpha}}{C_{L_\alpha}}, \quad (7)$$

expressed as a percentage of each design's declared c_{ref} [35]. Because body and propeller contributions are absent, it is a lifting-surface margin, not an installed neutral-point measurement. Rate derivatives were taken from the solver's quasi-steady stability-axis matrix on two meshes. The model provides neither $C_{m_{\dot{\alpha}}}$ nor installed control derivatives, and only centroid (parallel-axis) contributions to the inertia tensor could be assembled. Eigenvalues are therefore not reported. A reduced angle-of-attack/pitch-rate sign check, valid for any positive pitch inertia, is given in the Supplemental Material as a necessary-condition screen only.

F. Screening Criteria

Two exploratory thresholds were used to decide which candidates to carry forward: a surface-model static margin of at least 3% of c_{ref} in every declared loading state and a power reserve of at least 15% above the required shaft or engine power. Neither is an airworthiness standard or a quantified uncertainty allowance.

IV. Baseline Concepts and Audit (V0–V1)

A. Configurations

Figure 2 identifies the principal components of the three initial (V0) proposals. Astra proposed a tandem arrangement of two weight-supporting surfaces: a forward 12.0×2.0 m surface split into halves pivoted together for pitch and differentially for roll, and a rear 10.0×1.6 m surface fixed in flight. A seated operator, a proposed 72 kg, 18 kW motor, a long shaft driving a 2.4 m aft pusher propeller, and a rudder completed the concept. Fable proposed a braced biplane with 11.5×1.85 m decks separated by 1.85 m, outer upper-deck tip panels for roll, an all-moving horizontal tail, a fin and rudder, and two counter-rotating 2.3 m pusher propellers at $y = \pm 1.60$ m, a propulsion choice shared with the Wright Flyer. Opus proposed a biplane with 11×1.8 m decks, upper-wing balancing flaps for roll, a forward nacelle, a single 2.4 m pusher propeller, twin booms, and an aft elevator and rudder.

The V0 responses contained internal contradictions. Fable stated $x_G \approx 0.56$ m, whereas its own mass stations sum to approximately 1.14 m. In Opus, the declared shaft ended at $x = 2.0$ m while the propeller disk was at $x = 2.1$ m, and the main- and tail-skid heights did not define a common ground line. The V1 clarification replaced these statements with explicit mass ledgers and surface envelopes (Table 3). Figure 3 compares the V1 lifting surfaces with the Wright reconstruction. Astra’s rear surface is a second weight-supporting wing, not an elevator; Fable and Opus place a small surface aft of the biplane; the Wright canard lies ahead of it.

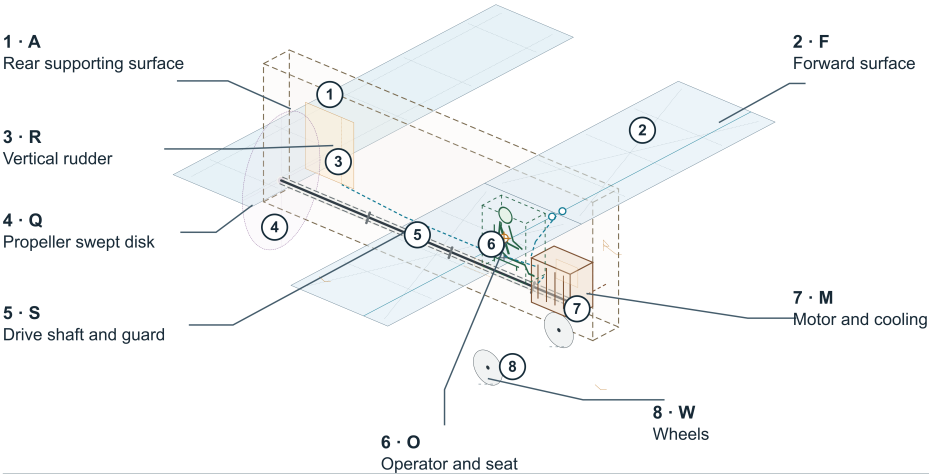
B. Lift Demand and Conditional Trim

At 12 m/s, the V1 designs require $C_L = 0.9173$ (Astra), 0.7730 (Fable), and 1.0475 (Opus) if vertical thrust is neglected. For comparison, a 340.2 kg (750 lb) loaded Flyer with a 518 ft² area requires 0.7860. With neutral controls at $\alpha = 4$ deg and 12 m/s, the lifting-surface models give $L - W \approx -1305$, -152 , and -1713 N, respectively. These single-angle values show lift demand only and cannot rank the designs.

An exploratory screen covered $\alpha = 0, 4$, and 8 deg, four added-drag coefficients ($\Delta C_D = 0, 0.05, 0.10$, and 0.20), and three added-drag arms ($a_D = -0.5, 0$, and 0.5 m). The 108 cases contain only 90 distinct combinations because a_D is irrelevant when $\Delta C_D = 0$, and the zero-increment cases omit all profile, wire, pilot, and gear drag. Of 95 interpolated control roots, all 24 Fable roots required the continuous tail to pass through the fixed fin; at the representative condition, the interpolated command was $\delta \approx -11.3$ deg. Nine Astra roots and 17 Opus roots required more power than the models’ unverified engine targets.

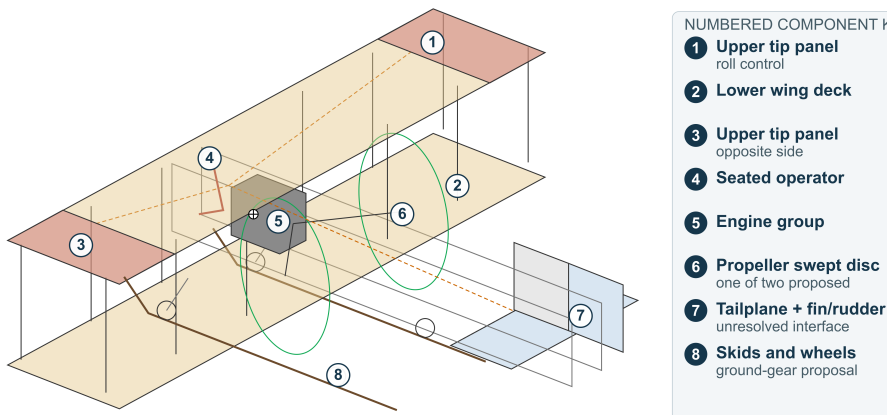
Table 3 reports the directly re-evaluated trim states at the representative condition, $\alpha = 4$ deg, $\Delta C_D = 0.05$, and $a_D = 0$. Both Astra surfaces lift: the forward and rear surfaces carry 2399.3 and 805.8 N, and their large opposing pitching moments nearly cancel. Opus’s lower and upper decks carry 1695.7 and 1903.3 N, while its tail carries a 121.5 N download. In both designs the aerodynamic pitching moment balances the thrust-line moment: $M_{\text{aero}} \approx -316$ N·m against $M_T \approx +317$ N·m for Astra, whose propeller hub lies 0.71 m below the CG, and +206 against -206 N·m for

Astra V0 — component geometry



a) Astra: tandem supporting surfaces, long-shaft pusher

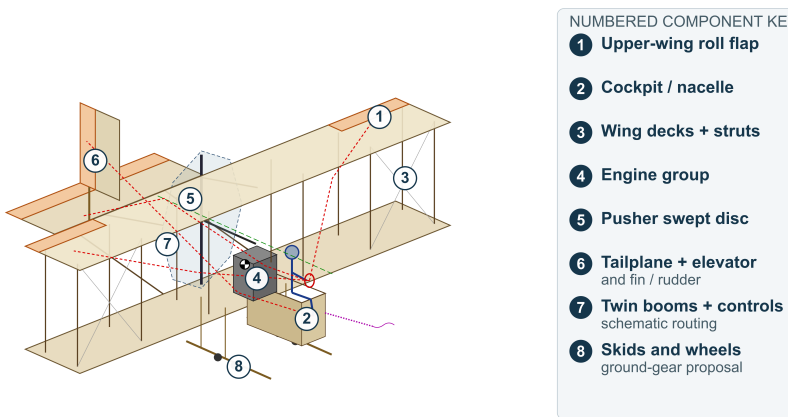
Fable V0: component identification



Source geometry preserved. Labels identify proposed assemblies, not validated installations.
V0 concept only: schematic drive and control paths; unresolved clearances and ground geometry.

b) Fable: braced biplane, all-moving tail, twin pushers

Opus V0: component identification



Source geometry preserved. Labels identify proposed assemblies, not validated installations.
V0 concept only: schematic drive and control paths; unresolved clearances and ground geometry.

c) Opus: biplane, single pusher, twin booms, aft tail

Fig. 2 Initial (V0) model-authored proposals with evaluator-added component identification; source geometry is unchanged. Dashed envelopes and control runs are schematic.

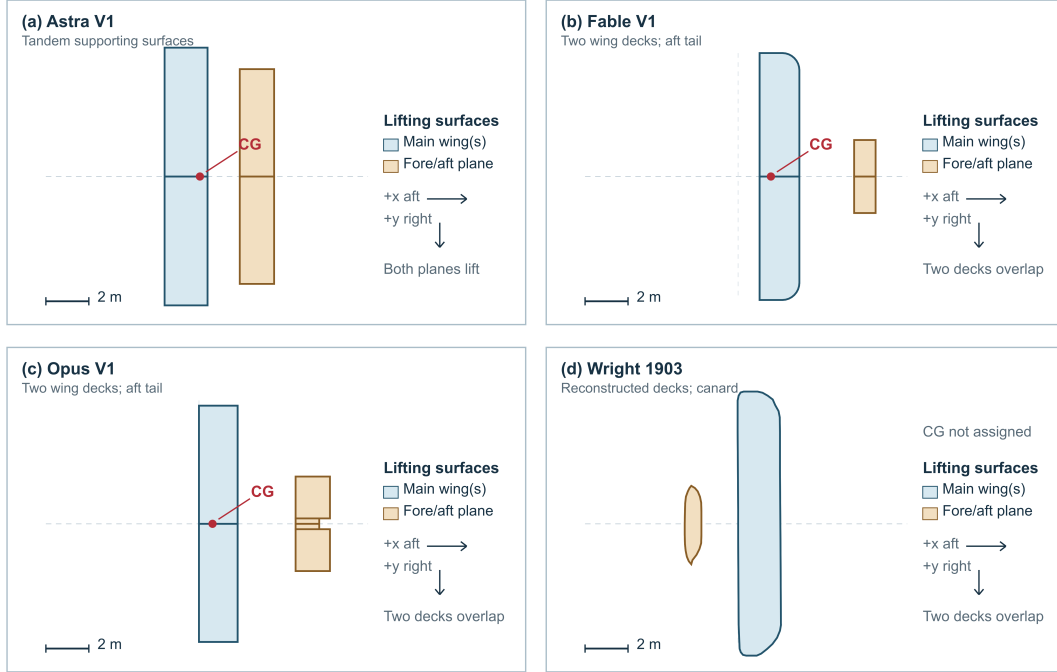


Fig. 3 Plan views of the V1 lifting-surface models and the Wright 1903 reconstruction, each in its own aft-positive frame. Biplane decks coincide in plan; markers show recomputed CGs.

Opus. Neither design was retrimmed with power off, so the pitch change that would follow a loss of thrust is not known.

The Opus result illustrates the sensitivity of a power verdict. Its required 13.35 kW exceeds the 13 kW target by about 3%, but the useful power is only about 8.04 kW, and a propulsive efficiency of 0.619 instead of the assumed 0.60 would remove the shortfall. Because ΔC_D and η_P are both assumptions, this gap identifies a sensitive requirement rather than a robust failure.

C. Static and Rate Derivatives

With controls fixed at the accepted trim states, Astra's pitching-moment residual increases from -18.1 to $+17.5$ N·m between $\alpha = 3.75$ and 4.25 deg, whereas Opus's decreases from $+9.9$ to -10.2 N·m. The corresponding margins are $SM = -9.94\%$ for Astra and $+5.14\%$ for Opus on the fine mesh, with coarse-mesh values within 0.2 percentage points. Astra's neutral point lies about 0.20 m ahead of its CG; the Opus neutral point lies about 0.09 m behind its CG. The quasi-steady pitch-damping derivatives are $C_{m_q} = -5.79$ for Astra and -6.59 for Opus. A negative C_{m_q} and the lift-pitch coupling can satisfy the reduced two-state sign check even for Astra's positive C_{m_α} , which shows that a static-margin verdict and a dynamic verdict must not be conflated; neither can be issued without inertia and unsteady derivatives. On this evidence, Opus V1 is the strongest initial proposal: it has a reconstructible ledger, an admissible trim, a restoring lifting-surface slope, and no known geometric conflict at its trim command, with a power margin that depends on assumed efficiency and drag.

Table 3 V1 mass properties and directly evaluated conditional trim at $\alpha = 4$ deg and $\Delta C_D = 0.05$

Quantity	Astra	Fable	Opus
m , kg	330	292	358
S , m ²	40.0	42.0	38.0
c_{ref} , m	2.00	1.85	1.80
x_G, z_G , m	0.044, 0.411	1.525, 0.752	0.632, 0.467
a_T , m	+0.711	−0.048	−0.433
δ , deg	0.042	−11.27 ^a	−1.909
V , m/s	15.46	12.83 ^a	16.86
T , N	445.2	—	476.4
P_{shaft} , kW	12.49	10.01 ^a	13.35
Engine target, kW ^b	18	not stated	13
SM , % c_{ref}	−9.94	—	+5.14

^a Interpolated root only; inadmissible because the tail intersects the fin.

^b Model-stated, unverified.

V. Feedback Iterations (V2–V13)

A. First Engineering Feedback (V2)

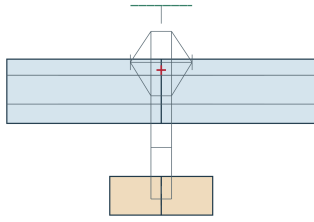
The V2 requests returned each model’s V1 results as labeled modern evidence and asked for revised geometry, mass ledger, controls, propulsion, and structure without dictating an architecture. Astra received its destabilizing slope and mechanism gaps; Fable, its tail–fin interference and missing power evidence; and Opus, its tail load, thrust-line moment, and power shortfall. Figure 4 shows the resulting three-view geometries.

The three responses differed in kind. Astra abandoned its tandem, variable-incidence layout for a fixed main wing, an aft horizontal tail with a separate elevator, outboard ailerons, a rudder, and a tractor propeller on a 0.7 m shaft. Fable replaced its continuous tail by two outboard halves at $|y| = 0.45\text{--}2.75$ m on a common torque tube hinged at $x = 5.65$ m, increased the declared tail area from 3.40 to 4.60 m², raised the fin, and moved the rudder hinge to $x = 6.60$ m; these changes directly target the V1 collision. Opus lengthened its skids, moved its booms and inner struts from $|y| = 1.40$ to 1.60 m to increase propeller clearance, raised the rudder, increased its mass from 358 to 360 kg, and proposed a slower operating point of 15 m/s.

Table 4 compares the independently retrimmed V2 states with V1. Astra’s margin changed sign, from −9.94 to +19.47%, but because wing, tail, controls, propulsion, and CG all changed together, the improvement cannot be attributed to any single design variable. Opus’s margin rose from 5.14 to 7.76%. At their different model-proposed speeds, the estimated engine demands are 8.46 and 11.67 kW for Astra and Opus, respectively, assuming $\eta_p = 0.55$ and 0.60 and $\eta_g = 0.95$; the total-drag assumptions differ from those of Table 3, so the two phases are not a same-mission comparison. Fable V2 could not be trimmed independently because the audit found an inconsistent tail-sweep envelope, a fuel centroid outside its stated box, unresolved drive and bracing routing, and no propeller pitch or speed.

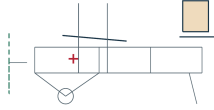
Astra | V2 | reference geometry / neutral controls

PLAN



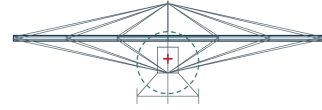
y (m) / x aft (m)
 2 m

SIDE



x aft (m) / z (m)
 2 m

FRONT



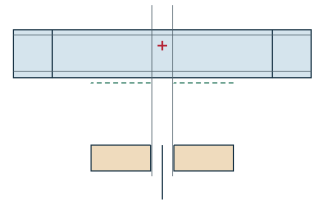
y (m) / z (m)
 2 m

Blue: wing | Ochre: tail | Dashed green: propeller disc | Red +: CG
 Coordinate reconstruction; not a construction drawing. Identical metric scale in all views; x aft, y right, z up.
 Declared frame, main-wing braces and gear shown; remaining joints, ribs, pilot and engine solids omitted.

a) Astra V2: fixed wing, aft tail, ailerons, tractor propeller

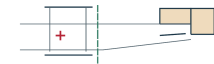
Fable | V2 | geometry unresolved

PLAN



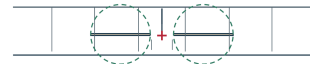
y (m) / x aft (m)
 2 m

SIDE



x aft (m) / z (m)
 2 m

FRONT



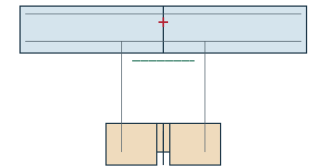
y (m) / z (m)
 2 m

Blue: wing | Ochre: tail | Dashed green: propeller disc | Red +: CG
 Coordinate reconstruction; not a construction drawing. Identical metric scale in all views; x aft, y right, z up.
 Fable: rectangular station envelopes; rounded wing tips are NOT reconstructed.
 Only explicit node chains drawn; ambiguous connectivity, drive conflicts and mass envelopes remain unresolved.

b) Fable V2: split all-moving tail, raised fin, twin pushers

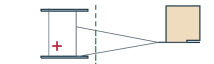
Opus | V2 | reference geometry / neutral controls

PLAN



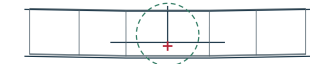
y (m) / x aft (m)
 2 m

SIDE



x aft (m) / z (m)
 2 m

FRONT



y (m) / z (m)
 2 m

Blue: wing | Ochre: tail | Dashed green: propeller disc | Red +: CG
 Coordinate reconstruction; not a construction drawing. Identical metric scale in all views; x aft, y right, z up.
 Only explicit node chains drawn; unspecified cross-bracing, landing gear, pilot and nacelle solids omitted.

c) Opus V2: lengthened skids, outboard booms, revised rudder

Fig. 4 V2 plan, side, and front views at neutral controls, rendered from the declared coordinates. Only specified surfaces and structural node chains are drawn.

Table 4 Independently trimmed lifting-surface states before (V1) and after (V2) the first feedback round

Quantity	Astra V1	Astra V2	Opus V1	Opus V2
m , kg	330	320	358	360
V , m/s	15.46	18.00	16.86	15.00
α , deg	4.000	-0.103	4.000	6.347
δ , deg	0.042	-0.330	-1.909	-3.168
Cm_α , rad^{-1}	+0.348	-0.813	-0.192	-0.286
SM , % c_{ref}	-9.94	+19.47	+5.14	+7.76
Cm_q	-5.79	-4.71	-6.59	-6.65
Power demand, kW ^a	12.49	8.46	13.35	11.67
Engine target, kW ^b	18	18	13	13

^a V1: shaft power with $\Delta C_D = 0.05$; V2: engine power with model-proposed drag, $\eta_g = 0.95$.

^b Model-stated, unverified.

B. Integration and Installation Rounds (V3–V13)

Phase D asked for integrated proposals with explicit mass, control, propulsion, and flight-dynamics plans, followed by targeted corrections. The V3 ledgers sum to 333.0, 309.5, and 361.3 kg for Astra, Fable, and Opus. Fable’s stated vertical-force and pitching-moment sums close arithmetically (E1), but its loads were its own estimates, and neither Astra nor Opus supplied a complete trim solution. Fable’s propeller subsystem studies (V3b and V5–V8-R) were not integrated into a complete aircraft.

The subsequent corrections exposed errors that a fluent narrative concealed. In V11b, Fable proposed seat stops and revised gear; its own 426 N drag estimate at 13 m/s and $\eta_p = 0.50$ require 11.08 kW at the shaft, or 12.74 kW with reserve, against a raised target of 12.8 kW that was not supported by an engine curve. Opus declined to change its design, but its own ledger contained a box whose stated extent, 0 to 0.5 m, was inconsistent with its stated center and length (−0.025 to 0.525 m). In V12–V13, Fable’s 26 component rows summed to 354.3 kg rather than the claimed 353.8 kg until a targeted correction. Under its heavy-case assumptions, Eq. (5) gives $P_{e,\min} = 20.05$ kW against a 19.5 kW target, a 0.52 kW shortfall at the shafts that would require $\eta_g = 0.977$ instead of 0.95 to close; coupling the model’s own specific engine mass of 5.1 kg/kW back into its drag raises the requirement to 20.20 kW and adds 3.5 kg. The same ledger implied 198.5 min of endurance rather than the stated 20 min. Opus V12 summed to 340.3 kg with a CG 0.081 m aft of its V3 value, so its earlier stability estimates could not be carried forward.

Rigid ground-contact, clearance, and inertia screens completed phase D. Fable’s static wheel reactions were positive in all eight declared loading states, but its smallest nominal installation gap was 0.010 m, between the fuel tank and the upper longeron, with no allowance for structural deflection. Opus’s declared tail-skid contact occurred at about 13 deg nose-up. Centroid-only inertia contributions were assembled for Fable and Opus as lower bounds; component intrinsic inertias were not available. Astra returned HOLD responses in V11b and V12 and supplied no adopted installed geometry.

VI. Refinement of the Fable Candidate (V14–V16)

A. Corrected Lifting-Surface Trim and Lateral Derivatives

The complete V15 Fable response specified a 353.2 kg aircraft with a 21 kW engine target. The other V15 responses remained on hold: Astra’s modeled propeller absorbed 5.4 kW more than its assumed shaft target, and Opus’s selected propeller fell 125.3 N short of its own drag estimate. Fable was therefore analyzed further as a power-off, rigid lifting-surface model comprising both wing decks, the split horizontal tail, and the fin and rudder.

An evaluator input error was found during this phase. The upper-wing center section and its movable tips had been assigned different AVL component identifiers, so the code treated them as aerodynamically separate bodies [31]. Correcting the identifier at unchanged geometry restored C_L from 0.747 to 0.797. The earlier result is retained in the Supplemental Material as a superseded evaluator error. In the corrected model, the all-moving tail is represented by rotating the normals over its full chord, and the fin and rudder share one component with aligned vortex stations.

At 13 m/s, the corrected fine-mesh trim is $\alpha = 5.860$ deg with a tail command of -4.068 deg (-8.068 deg absolute incidence) and a lifting-surface margin of only 0.97% (1.08% on the coarse mesh). In stability axes, $C_{n_\beta} = +0.0145 \text{ rad}^{-1}$ is restoring, $C_{l_\beta} \approx +0.0007 \text{ rad}^{-1}$ indicates essentially no dihedral effect, and the rate derivatives $C_{m_q} = -3.41$, $C_{l_p} = -0.364$, and $C_{n_r} = -0.0355$ indicate damping tendencies within the surface model. Of the six whole-aircraft equilibrium equations, only the vertical force and pitching moment were partially evaluated; side force, rolling moment, yawing moment, and axial force depend on unmodeled body, fin, propeller, and thrust terms.

B. Loading Envelope and Evaluator Tail Revision

The nominal margin did not represent the declared loading envelope. Re-trimming the two aft-CG states, with a 65 kg pilot at $x = 0.35$ m and 8 or 0 kg of fuel, gave $SM = -1.67$ and -1.58% on the fine mesh (Table 5). These states are in equilibrium but statically unstable, a direct example of the distinction between trim and stability.

To test whether the deficit could be closed, the evaluator scaled the exposed span of each tail half by a factor k while retaining the root station, chord, and hinge, so that the tail area is $4.60k \text{ m}^2$ and the assumed tail mass is $8k \text{ kg}$. On the coarse mesh, the light-pilot, full-fuel margin rose from -0.46% at $k = 1.15$ to $+4.15\%$ at $k = 1.75$. The $k = 1.75$ tail (8.05 m^2 , 9.15 m span, 14 kg) gave positive margins in all eight coarse-mesh loading states (4.12–9.71%) and 3.95 and 3.92% in the two critical fine-mesh states. This is an evaluator design intervention, not a model result.

C. Structural Mass Feedback

A larger tail is not a free aerodynamic gain. For a cantilever under unchanged line load, root bending moment scales with k^2 and tip deflection with k^4 , or factors of 3.06 and 9.38 at $k = 1.75$. Strip loads extracted at the trimmed states raised the root-moment magnitude for one tail half from 168.3 to 293.9 N·m. Using clear-wood properties for Sitka spruce at 12% moisture content ($E = 10.8 \text{ GPa}$, specific gravity 0.40, density 448 kg/m^3) [36], the lightest hollow-box

Table 5 Evolution of the Fable lifting-surface static margin at 13 m/s

Configuration (critical state)	Origin	m , kg	SM , % c_{ref}
V15, nominal loading	Model	353.2	+0.97
V15, aft CG, 8 kg fuel	Model	343.2	−1.67
V15, aft CG, no fuel	Model	335.2	−1.58
Tail $k = 1.75$, aft CG, 8 kg fuel	Evaluator	349.2	+3.95
Tail $k = 1.75$, aft CG, no fuel	Evaluator	341.2	+3.92
+ 2 kg point mass at hinge	Evaluator	343.2	+2.62
+ cantilever box spars ^a	Evaluator	348.0	−0.22
V16, eight loading states	Model ^b	347.1–375.1	+3.98 to +10.12

^a Coarse mesh; spars sized for a 20 mm tip-deflection screen.

^b Model response adopting and modifying the evaluator’s braced tail.

cantilever in a 210-section grid that met an illustrative 20 mm tip-deflection screen measured 160×160 mm with 3 mm walls. The two spars added 6.79 kg, which moved the CG aft enough to reduce the critical margin to -0.22% ; that branch was rejected. A 2 kg point mass at the hinge alone reduced the margin from 3.92 to 2.62%, and a fixed-neutral-point estimate placed the mass budget at only about 1.45 kg before the 3% screen was lost.

The evaluator therefore defined a tension-braced alternative (Fig. 5): two 120×100 mm boxes with 4 mm walls on the hinge line, a 40 mm central mast extending 0.35 m above and below that line, and four 4 mm tension-only wires to the outer tail stations. With zero wire prestress and the lower wire slack under the evaluated download, the predicted tip deflection is 11.49 mm, compared with 39.80 mm for the same box without bracing; the root moment is 139.2 N·m and the upper-wire tension 487.9 N. The boxes, wires, and mast weigh 8.43 kg, leaving 5.57 kg of the 14 kg tail allocation for ribs, covering, fittings, horns, and bearings. Beam-column effects, joint slip, load reversal, torsion, and flutter were not analyzed, and the mast–hinge mechanism is not yet reconciled with the fuselage structure.

D. Model Adoption and Loading Envelope (V16)

In V16, Fable adopted and modified the braced tail: it selected a continuous 100×80 mm box spar with 3.5 mm walls, two bracing masts at $y = \pm 0.55$ m, and upper and lower tie wires; moved the nominal pilot centroid forward to $x = 0.10$ m; and selected a 21.1 kW engine target. Its 31 ledger rows sum to 365.1 kg with $x_G = 1.5907$ m, within 0.13 mm of its reported value. Figure 6a shows the eight independently re-trimmed loading states. All margins are positive, from 3.98% in the light-pilot, aft-seat, no-fuel state to 10.12% in the heavy-pilot, forward-seat, no-fuel state; all tail incidences, from -1.64 to $+0.30$ deg, lie inside the declared travel of -16 to $+8$ deg; and the largest lift residual is 0.020 N. The audit also found that the reported tail-chord and mast envelopes were inconsistent with the stated rotation; correcting the centerline envelopes does not establish solid clearance.

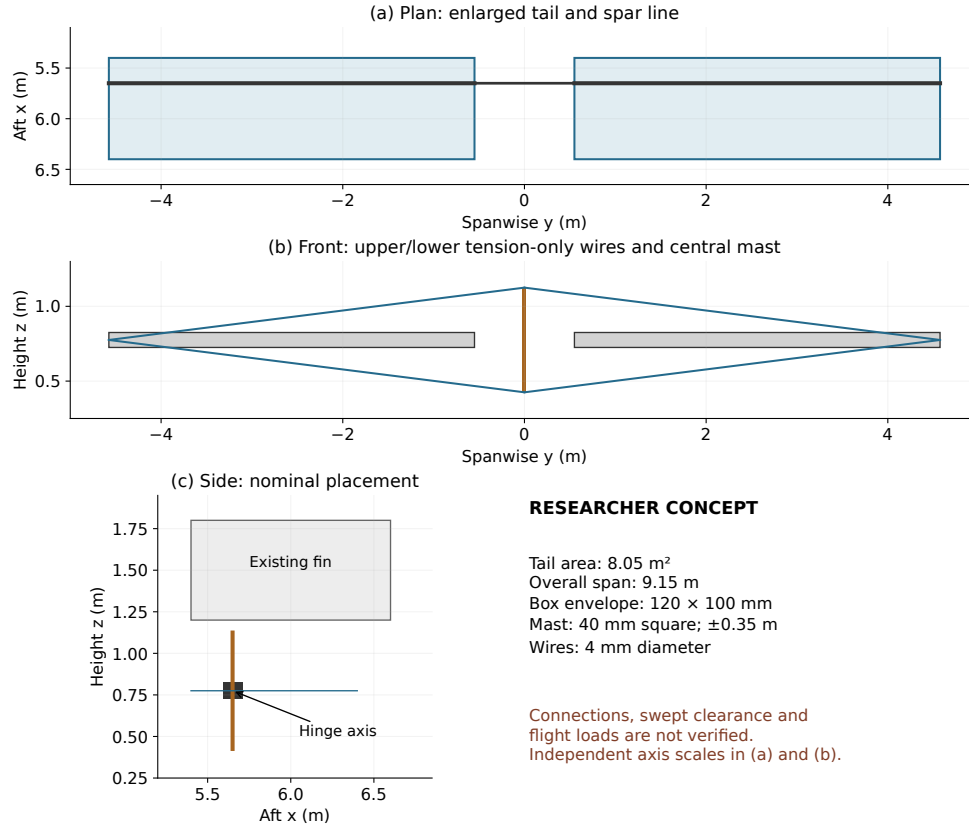


Fig. 5 Evaluator braced-tail concept for Fable: a) plan, b) front, and c) side views. Connections, swept clearance, and flight loads are not verified; a) and b) use independent axis scales.

E. Speed–Power Sensitivity

Figure 6b applies Eq. (6) to the heavy case (375.1 kg), for which the inferred efficiency is about 0.45 and the assumed available shaft power is $0.95 \times 21.1 = 20.05$ kW. At 13 m/s, replacing the model's induced-drag coefficient (0.0526) with the surface solution (0.0581) raises the shaft demand from 17.36 to 18.05 kW. Absolute headroom remains 1.99 kW, but the reserve falls to 11.0%, below the 15% screen. The nominal 365.1 kg case retains a 47% reserve (13.61 kW). At 12 m/s the heavy-case reserve is 22.5%, but the required $C_L = 0.993$ at $\alpha = 9.09$ deg has unknown stall margin; at 16 m/s the demand of 26.04 kW exceeds the assumed supply by 5.99 kW, although the lifting-surface equations still balance with margins of 10.85 and 8.71%, respectively. A larger engine would change mass, CG, and required lift, and that loop was not closed.

F. Final Status of Astra and Opus

Neither Astra nor Opus produced an installed geometry that could be trimmed after V2. For Astra V16, refining the model's single-cell propeller calculation to 128 radial cells, with the model's assumed polar and uniform induction retained, raised the predicted shaft demand at 15 m/s from 16.62 to 18.04 kW, 0.96 kW below its assumed 19 kW ceiling. For Opus V16, a 192-cell calculation at 13 m/s and 500 rpm gave 18.95 kW, above its 17.46 kW shaft target; an

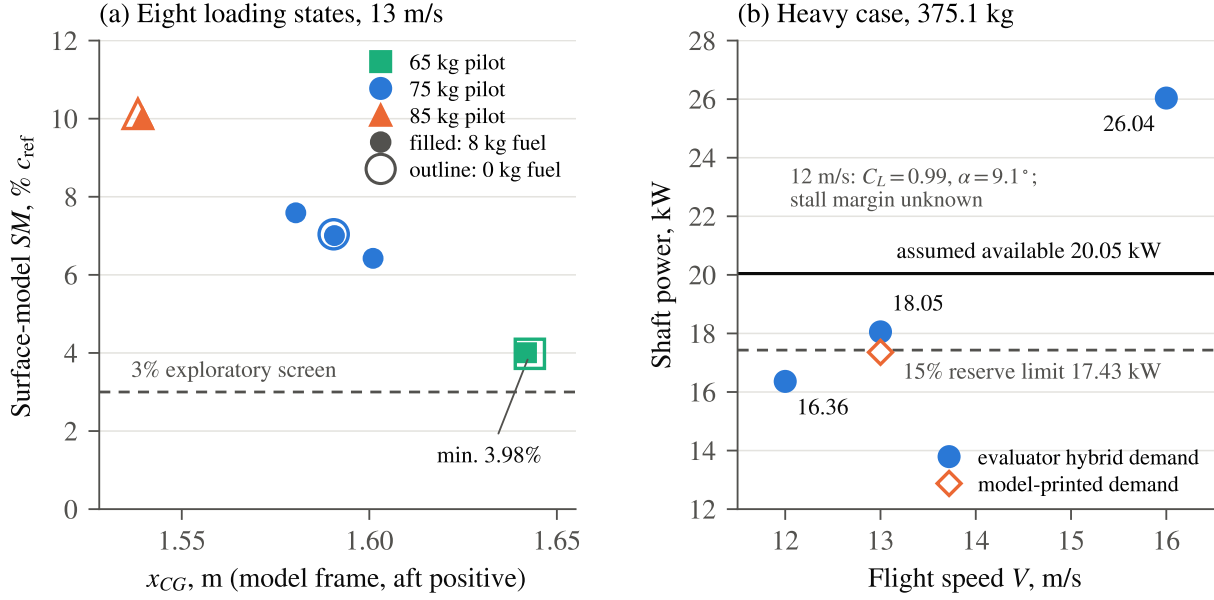


Fig. 6 Table V16 conditional results: a) lifting-surface static margin of eight independently re-trimmed loading states; b) heavy-case shaft power from Eq. (6) against the model’s assumed available power.

assumed constant crank torque gave an equilibrium at 484 rpm, 16.90 kW, and 897 N of thrust, and an assumed constant shaft power gave 488 rpm and 922 N. Neither engine characteristic is a measured map, and no same-speed drag estimate for the revised aircraft exists, so thrust sufficiency could not be assessed.

VII. Discussion

A. Evidence Reached by Each Design

Table 6 summarizes the highest evidence level reached by the latest version of each design, together with the Wright 1903 Flyer. Every generated design stops at E2, and every E2 result is conditional on rigid lifting surfaces, model-supplied sections, and assumed drag and propulsive efficiency. On the gates applied, Opus V1 was the strongest initial proposal, Astra V2 had the largest margin of the two independently retrimmed V2 designs, and Fable V16 is the only candidate evaluated across a loading envelope with lateral-directional derivatives. Fable’s positive margins, however, depend on a tail first proposed by the evaluator, and its power reserve depends on an efficiency inferred from its own printed values. None of these is a flightworthiness ranking.

B. Feedback and Design Change

The feedback record shows two productive patterns. When a defect was geometric or arithmetic, the models generally responded with a specific, testable change: Fable split its tail to clear the fin, corrected its ledger when the row sum was challenged, and added seat stops to bound its CG. When the information supplied was insufficient, as in the hash-only

Table 6 Highest evidence level reached by the latest version of each design

Gate	Astra (V2/V16)	Fable (V16)	Opus (V2/V12/V16)	Wright 1903
Mass and CG ledger	E1 (V2)	E1, 365.1 kg	E1, 340.3 kg (V12)	CG not sourced
Lifting-surface trim	E2 (V2)	E2, 8 loading states	E2 (V2)	Reconstruction only
Static margin, % c_{ref}	E2, +19.5	E2, +4.0 to +10.1	E2, +7.8	E4 (replica), about -25
Lateral-directional derivatives	Partial E2, no fin or body	E2, surface model	Partial E2, no fin or body	Published reconstructions
Power required vs. available	E2 propeller model; assumed ceiling	E2, 11% reserve (heavy)	E2 propeller model; drag unknown	E4, flown
Installed six-component trim	None	None	None	E4, flown
Inertia tensor	None	Centroid lower bound	Centroid lower bound	Published models
Dynamic modes	None	None	None	Published models
Structure	None	Tail concept only	None	Built and flown

V11a request, all three models declined to invent a design, which is the appropriate behavior. Three less productive patterns also recurred. First, a wholesale change of architecture, such as Astra’s conversion to a tractor monoplane with an aft tail, removed the defect without identifying its cause and reset most of the evidence. Second, power targets were raised on paper without an engine curve or propeller map, so apparent closure moved without new evidence. Third, arithmetic inconsistencies in ledgers, box extents, and endurance persisted across several rounds and were found only by recomputation. The evaluator also made an error, the component-identifier mistake of Sec. VI.A, which was detected by a consistency check. Independent verification is therefore required in both directions.

C. Comparison with the Wright Development

The Wright brothers’ development and the present experiment can be compared stage by stage, but the comparison is one of evidence classes rather than of silhouettes. The V0–V1 concepts correspond to the Wrights’ 1899–1900 position, when inherited tables were used to size a glider. The Wrights then flew gliders, found lift well below prediction, and measured their own section data [15, 16]; the V1 audit found trim conflicts and an unstable tandem, but with computation in place of measurement. Phase D corresponds to the 1903 integration of engine, propellers, transmission, and structure. The Wrights resolved that phase with their own engine and propeller theory and with flight [17, 37], whereas here the decisive inputs (an engine curve, an installed propeller map, total drag, and loaded clearances) remain assumptions. The 1904 and 1905 revisions changed camber, structure, ballast, canard size, vertical-surface area and moment arm, and rudder linkage, several of them within the 1905 season [23, 25, 26]; the Fable refinement in phase E addressed comparable loading and tail-sizing questions, but only in a rigid lifting-surface model.

Two further points qualify the comparison. First, the positive static margins of Astra V2, Opus V2, and Fable V16 do not make these designs “more stable” than the Flyer in any operational sense. The 1903 and 1905 Flyers were statically unstable and were flown because pilot action and control authority were adequate [23, 24]; canard geometry by itself establishes neither stability nor poor handling [38]. A meaningful comparison would require the generated

designs' dynamic modes and control authority, which in turn require the inertia and unsteady derivatives that are missing. Second, the brief's requirement of repeated directed flight and safe return is better matched by the 1905 Flyer III than by the 1903 machine, and the Flyer III was the product of two further years of flight testing. Some architectural similarities, such as Fable's twin counter-rotating pushers, may reflect recall of later aircraft; the models' rationales cannot reveal the causal origin of a choice, and an aft tail is not by itself anachronistic.

D. Limitations

The physical limitations are those of the evaluation model: total drag and installed propulsion are not validated; separated flow, stall, flexibility, and fabric deformation are excluded; inertia tensors are incomplete; and installed lateral-directional characteristics are unknown. The experimental limitations are equally important. Only three models were tested, each in a single sequence with unequal output budgets; feedback after V1 contained modern results; and the final Fable candidate incorporates an evaluator design. The Wright reconstruction is not matched to a sourced CG or a specific as-tested configuration. Handling-quality judgments, for example by the Cooper-Harper method [39], would require piloted simulation or test data not available here.

VIII. Conclusions

Three language models designed piloted, powered aircraft from a pre-1899 source packet, and their proposals were evaluated over a sequence of archived feedback rounds with an evidence-level protocol. The principal conclusions are as follows.

- 1) With clarification, the models produced auditable mass ledgers, surface envelopes, and control descriptions. Their unchecked statements nevertheless contained recurring arithmetic, geometric, and power-chain errors that only independent recomputation exposed.
- 2) The initial concepts failed different gates. Astra's tandem design trimmed but had a fixed-control static margin of -9.94% ; Fable's biplane could trim only with its tail passing through its fin; Opus's biplane trimmed with a margin of $+5.14\%$ but required about 3% more power than its unverified engine target under assumed drag and efficiency.
- 3) Feedback produced specific, testable changes when the defect was geometric or arithmetic, but it also produced wholesale re-architecture and paper increases in engine power that did not add evidence.
- 4) The final Fable candidate, which adopted an evaluator-proposed braced tail, has lifting-surface static margins of $3.98\text{--}10.12\%$ across eight loading states at 13 m/s. Its heavy-case power reserve falls from the model's apparent closure to 11.0% when the induced drag is computed, and its 12 and 16 m/s states are limited by unknown stall margin and a 5.99 kW power deficit, respectively.
- 5) No design reached installed six-component trim, accepted dynamic modes, or structural adequacy; all results are

at most independent, conditional computations. The Wright comparison raises rather than relaxes this standard, because the Flyers were statically unstable yet flown, and their success rested on measurements and flights that the present designs lack.

The next defensible steps are a component drag build-up with uncertainty, a measured or validated engine and propeller map, power-on and power-off six-component trim over a declared speed and CG envelope, complete inertia properties, unsteady derivatives, dynamic modes with control-authority analysis, and structural sizing of the load paths.

Appendix: Supplemental Material

The Supplemental Material contains the full V0 proposal plates and component keys; the Wright reconstruction three-view and sensitivity data; the V1 interpolated balances, surface loads, residuals, and pitch-response plot; the V2 force–moment diagrams, derivative screens, and reduced sign check; the V3–V13 ledger, power-chain, ground-contact, clearance, and inertia audits; the corrected and superseded V15 Fable results, including lateral and rate-derivative tables; the tail-span trade, strip-load, and structural-loop calculations; the V16 loading table and propeller audits; and a table of published Wright stability quantities.

Acknowledgments

The three language models named in Sec. II generated the evaluated designs; their outputs were treated as study material, not as technical evidence or as sources. An OpenAI coding assistant was used for research organization, analysis-code development, figure production, and drafting. Anthropic’s Claude was used to restructure and copy-edit the manuscript for this journal. The author reviewed all model-assisted text, calculations, citations, and figures and is responsible for the content. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Computations used the Unity Research Computing Platform at the University of Massachusetts Amherst.

Data Availability

The prompts, source packet, model responses (including truncated and failed attempts), versioned geometry, evaluator scripts, solver inputs and outputs, and checksum manifests are archived in a research repository (<https://github.com/Ehsan-Roohi/historical-aircraft-llm>). Records from V14 onward are held in the author’s archive and will be added to a fixed public release, with a persistent identifier, before publication. The AVL executable and copyrighted third-party material are not redistributed.

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